Remote Sensing Scene Classification via Pseudo-Category-Relation and Orthogonal Feature Learning

Jinsheng Ji, Xiankai Lu, *Member, IEEE*, Tao Zhang, *Member, IEEE*, Yiyou Guo, and Gongping Yang

***Abstract*— Remote sensing (RS) scene classification is a crucial component in the analysis of Earth observation data, aiding in a deeper understanding and monitoring of our dynamic planet. Its applications extend across various fields, including land management, urban analysis, and environmental monitoring. The complex semantic information in RS scene images and the relationships between different scene categories present significant challenges to improving scene classification tasks. Unlike previous methods that only focus on network structure or feature encoding, our approach emphasizes the association of scene categories, integrating feature learning and knowledge transfer together to enhance the analysis of scenes at a higher semantic level. To this end, we propose an RS scene classification scheme based on pseudo-scene category-relation reasoning and orthogonal feature (OF) learning modules, capturing the inherent semantic connections among diverse scene classes. Additionally, we introduce cascaded attention (CA) and selected separation modules to strategically optimize the network, targeting challenging classes with high feature similarities. Knowledge is then distilled across different branches, guiding to enhance the model's robustness and prediction accuracy. Experiments are conducted on three challenging RS scene datasets of AID30, UCMerced21, and NWPU-RESISC45 to validate the effectiveness of the learned pseudo-category relationships. The results demonstrate that the proposed framework outperforms existing hierarchical approaches in leveraging the hierarchical structure of RS scene images.**

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***Index Terms*— Knowledge distillation, orthogonal feature (OF) learning, pseudo relation learning, remote sensing (RS), scene classification.**

I. Introduction

SCENE classification represents one of the most extensively explored domains within the field of parsing Earth observation data, offering a powerful means of comprehending and monitoring the dynamic environment of our planet. Numerous critical applications, including land management [1], urban analysis [2], environmental monitoring, disaster management, and agricultural assessment, hinge on the efficacy of remote sensing (RS) classification models. Nevertheless, RS scene classification is a task that still presents many challenges [3], [4]. On one hand, RS imagery is acquired under different environmental and imaging conditions, giving rise to considerable variations that increases the difficulty in obtaining robust classification. Objects and land cover classes can exhibit substantial visual disparities across diverse conditions. On the other hand, obtaining deep understanding of the semantic content and context of the scenes is also crucial to achieve accurate and reliable classification of RS scenes [4], [5], [6], [7], [8].

In the field of RS scene classification, each scene image can be associated with one or more category labels, a feature derived from both the semantic context of the image and the specific objects it contains. This stands in contrast with the conventional paradigm of single-label classification, where each image is exclusively assigned to a single class. The adoption of multilabel classification in this context allows for a more detailed understanding and accommodates the complexity and diversity inherent in RS imagery [9], [10], [11], [12], [13]. These category labels can be generated in two fundamental ways. First, they can be determined entirely based on the visual content of objects within the image. Alternatively, a more intricate approach involves establishing hierarchical label relationships, presenting a structured semantic hierarchy [14]. This hierarchical label framework, distinct from general multilabel systems, holds the potential to encode hierarchical semantic relationships across different RS scene images [15], [16]. This approach surpasses traditional multilabel classification by providing valuable insights into the hierarchical nature of the content, thereby fostering richer and more contextually aware scene classification.

To harness the inherent semantic relationship among different scene categories more effectively, numerous research

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efforts have emerged [14], [17], with approaches ranging from the design of different network structures to the utilization of multitask learning to attain more accurate prediction results. Unlike traditional single-task models, multitask learning leverages multiple interrelated tasks to enhance overall classification performance [18], [19], [20]. By jointly training a model on these tasks, multitask learning enhances the model's ability to understand complex RS scenes. Despite the potential offered by multilabel information to optimize classifiers within the multitask learning framework, there remains room for exploration in uncovering informative semantic correlations among various scene categories to distinguish RS scene classes more effectively.

Knowledge distillation has emerged as a promising technique for addressing the problem of learning efficient and distinctive features from the RS images. This process entails the transmission of knowledge from a sophisticated teacher model to a lightweight student model. It empowers the student model to achieve comparable performance while consuming fewer computational resources [21], [22], [23], [24], [25]. Through harnessing the distilled knowledge learned by the teacher model, the student model acquires the ability to generalize effectively and make accurate classifications of RS scenes.The knowledge distillation process involves utilizing the teacher model's outputs as soft targets for training the student model. These soft targets furnish additional insights into the relationships among different classes, enabling the student model to capture discriminative details and make more accurate predictions [26], [27], [28], [29].

Inspired by the success of soft-label and knowledge distillation techniques [22], [30], [31], which involve transferring acquired knowledge from a teacher model to a student model, this article presents a novel RS scene classification scheme that employs pseudo-category-relation learning and hierarchical knowledge fusion to address the task of RS scene classification. Specifically, the scene classification datasets are structured in accordance with standard land cover categorization, and each image is assigned three-level labels in a hierarchical format. Subsequently, a hierarchical information fusion module is designed to exploit the inherent semantic relationships within the RS scene datasets. The acquired knowledge is then distilled and transformed into soft labels, serving as guidance for the lower branch-network to acquire more informative features. Furthermore, an innovative selected separation loss is proposed, focusing on the challenging classes with high feature similarities.

We summarize the contributions of this work as follows.

1) We design a pseudo-category-relation learning module to abstract and understand the semantic relationships between scene categories, providing guidance for handling difficult categories.
2) We introduce a novel hierarchical knowledge distillation and fusion module to manage the challenging categories identified by the category-relation learning module.
3) We propose innovative cascaded attention (CA) and orthogonal feature (OF) learning modules, utilizing the learned relationships to optimize the network for informative feature learning.
4) We manually organize the RS scene datasets with three-level hierarchical labels, offering a new perspective for analyzing and evaluating scene classification tasks.

The structure of this article is as follows. Section III provides an overview of prior research RS scene classification. In Section IV, we offer a comprehensive description of our proposed method. Section V presents the experimental results. In Section VI, we summarize and conclude this article.

II. Related Works

A. Hierarchical Image Classification

RS data, including satellite imagery and aerial photographs, provide valuable insights into land surfaces and their dynamic changes. However, the complex nature of RS scenes makes traditional single-label classification methods challenging where multiple objects and diverse land cover types often appear within a single image. Over the years, researchers have made significant advancements in addressing the challenges of multilabel RS scene classification. Instead of assigning a single label to an image, hierarchical label image classification organizes labels hierarchically, offering a more detailed and adaptable representation of visual content. This approach proves especially valuable when dealing with intricate and diverse images that may contain multiple objects or exhibit varying levels of detail. Numerous techniques and methodologies have been developed, emphasizing improvements in classification accuracy, feature representation, model architectures, and dataset construction.

In [14], two-level labels for each scene image are organized and hierarchical label information is leveraged to enhance classification accuracy by incorporating attention mechanisms and multitask learning. Li et al. [32] introduce a multitask–multistructure fusion model designed to incorporate various label structures, thereby enhancing hierarchical classification. In order to leverage multilabel information, the work of Kolisnik et al. [33] proposes a teacher-forcing training algorithm, wherein the lower level branches are trained using the real class labels derived from higher level classes, rather than solely depending on predicted labels. Dhall et al. [34] propose a hierarchical image classification framework by incorporating label-hierarchy knowledge into an arbitrary CNN-based classifier, ultimately improving overall performance. They employ labels from coarse to fine to jointly optimize the classification network, fostering a better understanding of objects in images. Distinct from these prior methods, our approach integrates hierarchical outputs into an embedded format for joint classifier training and results of hierarchical levels are fused together to achieve more accurate predictions.

B. Knowledge Distillation for Scene Classification

Knowledge distillation is a technique for transferring insights from a complex teacher model to a smaller student model, enabling the student to achieve performance comparable to the teacher. By leveraging the knowledge distilled from the teacher, the student model can learn more generalized and robust features, enhancing its accuracy and robustness in classifying RS images. Recent advancements have introduced

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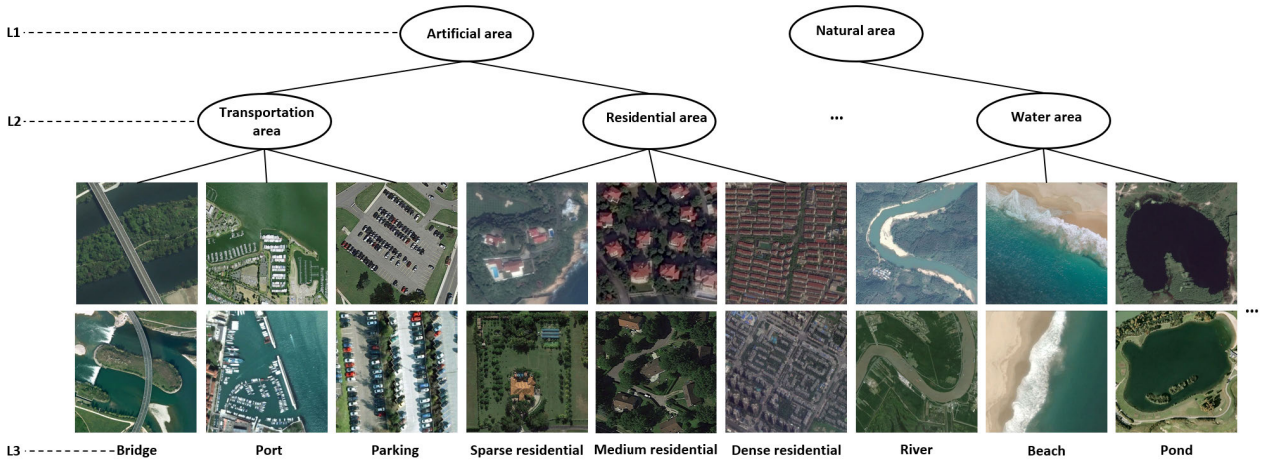


Fig. 1. Example images of UCM21, AID30, and NWPU45 datasets with three-level labels. Different scene categories are further organized into two upper levels according to their semantic information, such as "transportation," "water area," and "residential area."

TABLE I

ORGANIZED HIERARCHICAL RS SCENE CLASSIFICATION DATASETS WITH THREE LEVELS BASED ON UCM21, AID30, AND NWPU45 DATASETS

		Class number		
Dataset	**Sample Num**	**L1**	**L2**	**L3**
UCM21 [37]	2,100	2	8	21
AID30 [38]	10,000	2	11	30
NWPU45 [39]	31,500	2	11	45

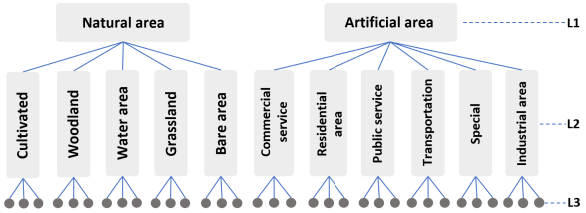


Fig. 2. Illustration of constructed three-level hierarchy for RS scene categories.

various knowledge distillation methods specifically designed for the task of RS scene classification. For instance, a novel framework incorporating adaptive reinforcement learning is proposed in [21]. This framework includes a multiscale core feature imitation module, which facilitates effective knowledge transfer by adaptively selecting core features and emphasizing the representation of smaller objects.

To achieve a balance between convolutional neural network accuracy and the number of model parameters, a novel dual knowledge distillation model is proposed in [28]. This model integrates dual attention and spatial structure to improve model efficiency without compromising performance. In [35], a novel pair-wise similarity knowledge distillation method that effectively reduces model complexity while maintaining high accuracy and efficient deep neural network is proposed to handle RS image scene classification. Furthermore, Xing et al. [36] utilize a collaborative consistent knowledge distillation framework to enhance classification accuracy of RS images on embedded devices. In contrast, our knowledge distillation framework relies on the hierarchically embedded results from the teacher network. Moreover, the predicted results, which incorporate hierarchical labels, serve as a guide for optimizing the student network to focus on distinguishing classes that pose challenges for the teacher network. In our hierarchical knowledge distillation model, the student network can leverage knowledge from the teacher network to achieve even better performance.

III. HIERARCHICAL LABELS CONSTRUCTION

Generally, scene image categories are obtained based on human subjective consciousness. On this basis, these hierarchy of the scene categories can be further obtained according to their semantic information as demonstrated in Fig. 1. We follow up the land category standard and refer the two-level labels in [14] to construct the hierarchical labels. In addition to the categories at the finest level (i.e., L_3), we also incorporate two upper level category groupings, as illustrated in Fig. 2. This hierarchical organization allows scene categories to be structured according to a tree-based hierarchy, facilitating a more intuitive understanding of their relationships. At the upper levels, the categories are divided into two main groups: **natural scenes** and **artificial scenes**. The **natural scenes** group includes categories such as cultivated areas, woodlands, grasslands, water areas, and bare areas, all of which represent environments shaped primarily by natural processes. These categories are distinguished by features such as vegetation, water bodies, and exposed soil, reflecting the diversity of Earth's natural landscapes. The **artificial scenes** group, on the other hand, encompasses categories such as commercial service areas, industrial zones, residential neighborhoods, public service spaces, special-purpose areas, and transportation networks. These categories are characterized by human-made structures and landscapes, highlighting the influence of human activities on the environment.

The motivation of grouping these scenes into two upper level categories are based on the content and semantic information of the scene images. To be specific, the artificial scenes are characterized by human-made structures and landscapes, while natural scenes are untouched by human activity and represent the Earth's natural diversity. Both types of scenes have distinct characteristics that can effectively

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describe and represent the scene images. Following this principle, the cultivated is grouped into the class of natural scene as its overall state of the scene still tends to be natural and some typical contents of artificial scene (i.e., buildings) are not included. Compared with AID30 [38] and NWPU45 [39], some classes are not included in the UCMerced Land Use dataset [37] and images are organized into eight parental classes belonging to level L_2 as in Table I.

A. UCMerced Dataset

It is a widely recognized benchmark dataset in the field of RS scene classification [37]. Originating from the University of California, Merced, this dataset offers a rich and varied collection of aerial imagery spanning 21 distinct land use classes. In this article, we denote it as UCM21. These classes encompass a wide range of land cover categories, including agricultural fields, forests, residential areas, highways, airports, and several other distinct scene categories. This diversity reflects the complexity and richness of real-world land cover, making the dataset highly valuable for RS research. In total, the dataset comprises 2100 images, with each class represented by approximately 100 images. The images are standardized to a resolution of 256×256 pixels and were captured under the purview of the National Agricultural Imagery Program (NAIP). Notably, the dataset encompasses diverse environmental and imaging conditions, rendering it a robust choice for evaluating the efficacy and adaptability of RS scene classification models.

B. Aerial Image Dataset

The Aerial Image Dataset stands as a comprehensive and highly regarded benchmark dataset constructed for the analysis of RS images [38]. In this article, we denote it as AID30. This dataset serves as a valuable resource for fostering research across various RS tasks, encompassing activities such as scene classification and object detection. This dataset is notably composed of 10 000 aerial images, spanning a total of 30 distinct land use categories. These categories span a broad spectrum, incorporating urban areas, farmland, forests, airports, water bodies, and many others. Each image maintains a fixed resolution of 600×600 pixels, sourced from diverse geographical locations and captured under varying imaging conditions. This heterogeneity in data sources equips researchers with a versatile array of variations and challenges for their investigations.

C. NWPU-RESISC45 Dataset

The dataset [39], referred to as NWPU45 in this article, is a widely used benchmark dataset. It consists of 31 500 images covering 45 different land use categories. The dataset includes a diverse collection of RS images captured from various sources, such as satellite imagery and aerial photographs. Each image has a standardized resolution of 256×256 pixels and represents a distinct land use category, such as airports, farmlands, forests, rivers, and urban areas.

D. Hierarchical Label Construction

In order to describe the constructed hierarchical relationships of these categories, the category-relation matrix $\bar{\boldsymbol{D}}$ for describing the semantic distance between different scene categories across the hierarchy is introduced as shown in the following equation:

$$\bar{\boldsymbol{D}} = \begin{bmatrix} w_{1,1} & \cdots & w_{1,c} \\ \vdots & \ddots & \vdots \\ w_{c,1} & \cdots & w_{c,c} \end{bmatrix} \tag{1}$$

where $w_{i,j} \in \{0, 1, 2\}$ is based on the constructed label hierarchy. When $w_{i,j} = 0$, it indicates that classes i and j share the same parental node, implying a close semantic distance and making them hard to distinguish. If $w_{i,j} = 1$, it means that classes i and j do not share the same parental node, but their parental nodes share a common parent. Finally, $b_{i,j} = 2$ signifies a large semantic distance between classes i and j, as they do not share the same parental node in the constructed hierarchy. Here, c represents the number of classes. Essentially, this hierarchical information is organized based on the semantic information of the scenes, containing rich insights about the RS scene category relationships, and providing valuable guidance for the scene classification model.

IV. Proposed Method

An overview of the proposed RS scene classification scheme is shown in Fig. 3. To illustrate the learning and fusion mechanisms in our framework, we first present the details of pseudo-category-relation learning. The learned category-related matrix contains rich semantic information that describes the relationships among different scene categories. We further embed this learned matrix into the network optimization process, allowing the three-branch models to collaborate effectively in handling these challenging categories.

A. Pseudo-Category-Relation Learning

Consider an RS scene classification task involving c classes. The input data, denoted as $\boldsymbol{X} = \{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n\}$, where n is the number of samples, and each sample $\boldsymbol{x}_k \in \mathbb{R}^{3\times w\times h}$ has dimensions of width w and height h. The standard approach for completing the classification task is to learn the function $\mathbf{f}_\theta : \boldsymbol{\mu}_k \rightarrow y_k$ by performing gradient descent on a loss function defined over the extracted feature $\boldsymbol{\mu}_k$ from input image data $\boldsymbol{x}_k$ and the labels $y_k \in \{1, \ldots, c\}$. Informative features $\boldsymbol{\mu}_k$ can be further utilized to evaluate the similarity between different categories. In this work, we propose a framework that measures multilevel semantic information between RS scene categories to better handle difficult-to-distinguish categories. The matrix $\boldsymbol{D}_l \in \mathbb{R}^{c\times c}$, which measures the similarity of different categories, is defined as in the following equation:

$$\boldsymbol{D}_l = \begin{bmatrix} d_{l,1,1} & \cdots & d_{l,1,c} \\ \vdots & \ddots & \vdots \\ d_{l,c,1} & \cdots & d_{l,c,c} \end{bmatrix} \tag{2}$$

where $d_{l,i,j} = \mathbb{E}[\|\boldsymbol{\mu}_{l,i,m} - \boldsymbol{\mu}_{l,j,n}\|^2]$ is calculated to represent the feature distance between class i and j at the lth level

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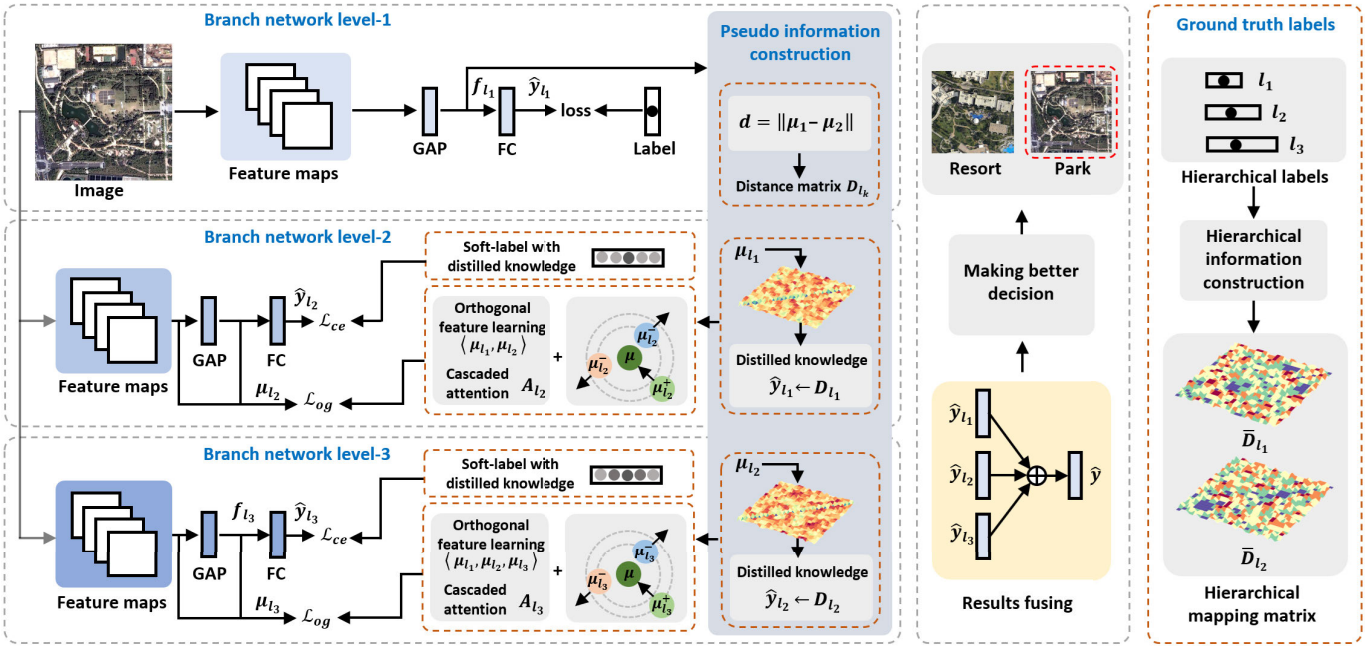


Fig. 3. Illustration of the proposed pseudo-category relation and OF learning for RS scene classification. The learned semantic information regarding to the scene categories is then further distilled and transformed into soft label and CA to guide the branch-networks to learn robust and reliable results. The manually labeled hierarchical labels can also be incorporated into the proposed RS scene classification scheme.

branch network, i.e., $l \in \{1, 2, 3\}$. Here, m and n denote different samples in each class. c is the class number. $\mathbb{E}$ denotes the averaged feature distance across the overall feature-domain learned by the lth branch network.

The rationale behind designing multiple-level networks is to employ them to partition and collaborate on the RS scene classification tasks. Specifically, the network at the lower level allocates more attention to categories that are challenging to distinguish at the upper level. Following this approach, we integrate the calculated semantic information from the lth level and its upper levels to identify those classes that still exhibit significant margin distances in the feature domain. The fused matrix $\hat{\boldsymbol{D}}_l$ serves as guidance for the optimizer to further concentrate on these challenging categories, which have not been adequately distinguished by their upper level branch networks. Further elaboration can be found in the accompanying in the following equation:

$$\hat{\boldsymbol{D}}_l = \mathbb{E}[\boldsymbol{D}_l] = \begin{bmatrix} \mathbb{E}^l_{j=1}[d_{j,1,1}] & \cdots & \mathbb{E}^l_{j=1}[d_{j,1,c}] \\ \vdots & \ddots & \vdots \\ \mathbb{E}^l_{j=1}[d_{j,c,1}] & \cdots & \mathbb{E}^l_{j=1}[d_{j,c,c}] \end{bmatrix} \tag{3}$$

where the learned semantic information from the current level l and its upper levels (i.e., $\boldsymbol{D}_1, \ldots, \boldsymbol{D}_l$) is summed together to obtain the averaged result $\hat{\boldsymbol{D}}_l$ for lth branch network. For the $\bar{\boldsymbol{D}}$ generated based on the ground truth hierarchical labels, the same category-relation matrix $\bar{\boldsymbol{D}}$ will be applied to all three branch networks. That is, $\boldsymbol{D}_l = \bar{\boldsymbol{D}}$ where $l = 1, 2, 3$. Compared to the learned $\boldsymbol{D}_l$, the fixed $\bar{\boldsymbol{D}}$ based on the ground truth hierarchical labels provides some semantic information about category relationships. However, because it is fixed, it does not adapt or match as well to the tasks and models as the learned $\boldsymbol{D}_l$.

We further normalize the feature distance d and get $\hat{d}_{l,i,j} \in [0, 1]$ for measuring the similarity between class i and class j. Each element $\hat{d}_{l,i,j}$ reflects the distinguishing ability of the lth branch model for class i and j. Specifically, $\hat{d}_{l,i,j} > t$ indicates that the model can achieve a margin greater than the threshold t, resulting in good classification performance for distinguishing samples belonging to class i and j. Conversely, if $\hat{d}_{k,i,j} < t$, it suggests that samples of class i and j are easily confused with each other. To convert this high-level semantic information into supervisory information similar to category labels, we extract the hth row of relation matrix $\hat{\boldsymbol{D}}_{k,h,:}$ to calculate the relation vector for class h by the following equation:

$$\hat{\boldsymbol{g}}_{l,h} = 1 - \hat{\boldsymbol{D}}_{l,h,:} \tag{4}$$

where $\hat{\boldsymbol{g}}_{l,h}$ is a vector that measures the relations between class h and all c classes (including itself). Ideally, samples from different categories should be well distinguished. This means that each element $\hat{\boldsymbol{g}}_{l,h,h} \to 1$ for evaluating the distance between class h and itself at the lth level, and the corresponding distances with other categories $\hat{\boldsymbol{g}}_{k,h,j} \to 0$ for $h \neq j$. In this ideal scenario, it corresponds to the original one-hot category labels $\boldsymbol{y} = \{0, \ldots, 1, \ldots, 0\}$.

B. Hierarchical Knowledge Distillation and Fusion

It should be noted that the calculated category relation result $\hat{\boldsymbol{g}}_{l,h} \in \mathbb{R}^c$ contains rich semantic information about the relationships between different scene categories at the lth level. Each element $\hat{\boldsymbol{g}}_{l,h,j}$ in $\hat{\boldsymbol{g}}_{l,h}$ can be regarded as a soft value, i.e., $\hat{\boldsymbol{g}}_{l,h,j} \in [0, 1]$, instead of the binary value of $\{0, 1\}$ as the general one-hot label, for evaluating the relation between class h and j. We integrate this valuable information into the model to address these challenging scene categories. To achieve this, the learned semantic information from the lth level branch network is applied to the next lower level branch network, i.e., the $l + 1$ level, optimizing it to distinguish the difficult

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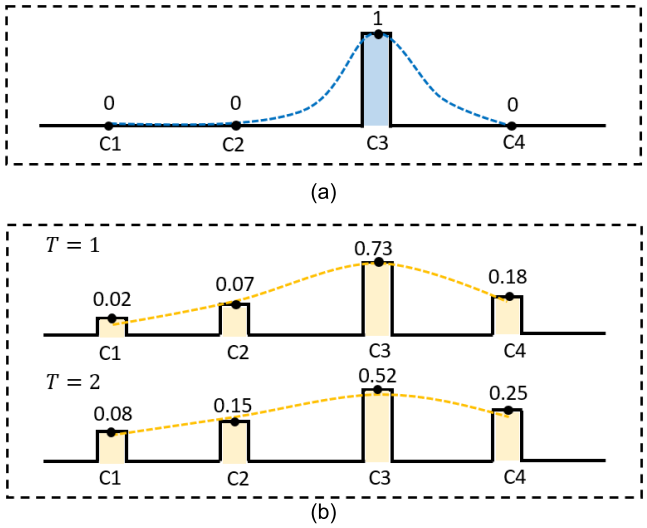


Fig. 4. Illustration of one-hot label and the knowledge distilled soft label. Higher temperatures of T bring flatter soft labels. (a) One-hot label. (b) Knowledge distilled soft-label.

classes that have not yet been correctly classified by the lth branch network.

Following the suggestion in [40], incorporating soft labels during the training of a classification model can provide enhanced regularization, particularly when dealing with noisy annotations. Soft labels often result in more flexible decision boundaries compared to fixed one-hot labels, potentially enhancing generalization performance. Additionally, soft labels provide dynamic regularization during model optimization, as illustrated in Fig. 4. This soft label provides more fine-grained information about the relationship between RS scene categories. By integrating soft labels that incorporate category relationship information, the robustness of the classifier is enhanced. This approach also enables each branch network to effectively account for the relationships between different categories in a more targeted manner. The detailed definition of generating soft labels based on the calculated relation vector $\hat{\boldsymbol{g}}_{l,h}$ for class h at the lth level can be found in the following equation:

$$\hat{\boldsymbol{p}}_l(\cdot|h) = \frac{e^{(\hat{g}_{l,h}/T)}}{\sum_j^c e^{(\hat{g}_{l,h,j}/T)}} \tag{5}$$

where $\hat{g}_{l,h}$ is the normalized relation weights of class h with other classes at the lth level branch network. These weights are further transformed into the soft label according to the temperature $T = 2$. c is the total class number.

It can be noted that the each element in the soft-label $\hat{\boldsymbol{p}}_l(\cdot|h) = \{\hat{p}_l(1|h), \ldots, \hat{p}_l(c|h)\}$ represents the relation between two corresponding categories. Generally, the values of $\hat{p}_l(j|h)$ denote the learned relations between class j and h. Here, $\hat{p}_l(j|h) = 1$ when $j = h$, similar to the function of the ground truth label, indicating that samples belonging to the same class should have high relations in the feature space. The category-relation information learned by the upper level branch network acts as the soft-label to guide the lower level branch network in learning a robust classifier. The detailed

Algorithm 1 Proposed hierarchical information fusion and knowledge distillation algorithm for RS scene classification

Input: Image set $\{\boldsymbol{X} \in \mathbb{R}^{3\times h\times w}\}$ and labels $\{\boldsymbol{y}\}$;
Output: Classification model, category-relation matrix, and predicted results;
Initialize relation matrix $\boldsymbol{D}_{l_k} \leftarrow 0$;
for *epoch* $e = 1$ *to* E **do**
  if $\boldsymbol{y}$ *contains hierarchical information* **then**
    $d_{k,i,j} \leftarrow \{0, 1\}$ according to $\boldsymbol{y}$;
  else
    $d_{k,i,j} \leftarrow \sum \|\boldsymbol{\mu}_{k,i,m} - \mu_{k,j,m}\|^2$;
  end
  $\hat{\boldsymbol{D}}_k \leftarrow \sum_{j=1}^{k} \hat{\boldsymbol{D}}_j$ by Eq. (3);
  Normalize $\hat{\boldsymbol{D}}_k$ and get $\hat{\boldsymbol{g}}_{k,h}$ by Eq. (4);
  Generate soft-label $\hat{\boldsymbol{p}}_k(\cdot|h)$ by Eq. (5);
  Calculate classification loss $\mathcal{L}_{k,cl}$ by Eq. (6);
  Optimize by letting $0 \leftarrow \frac{1}{2}\sum_{j=1}^{k-1}(\hat{\boldsymbol{\mu}}_{l_j}^T \hat{\boldsymbol{\mu}}_{l_k})^2$;
  Calculate CA $\hat{\boldsymbol{A}}_{l_k} \leftarrow \prod_{j=1}^{k} \boldsymbol{A}_{l_j}$;
  Calculate $\mathcal{L}_{k,sp} \leftarrow f(\boldsymbol{\mu}, \boldsymbol{\mu}^+, \boldsymbol{\mu}^-)$;
  Total loss $\mathcal{L} \leftarrow \mathcal{L}_{l_k,cl} + \mathcal{L}_{l_k,og} + \mathcal{L}_{l_k,sp}$;
  Back gradient propagation and update the network;
end

calculation is defined in the following equation:

$$\mathcal{L}_{l,\mathrm{cl}} = -\frac{1}{c}\sum_{j=1}^{c}(y_{l,j} + \hat{p}_{l,j}) \log \hat{y}_{l,j} \tag{6}$$

where $y_{l,j}$ and $\hat{p}_{l,j}$ are the one-hot label and soft label for class j at the lth level. c denotes the class number and $\hat{y}_{l,j}$ is the predicted output. By leveraging the distilled insights from the upper level branch network, the lower level branch network can be optimized to enhance robust regularization and improve the performance of recognizing various RS images. Cooperating with the soft-label distilled from the upper level branch network, the category-relation-based separation loss is proposed to further enhance the ability of distinguishing similar classes.

Different from general triplet loss, the selected separation pairs are constructed according to the pseudo-category-relation matrix. The separation loss is formulated as

$$\mathcal{L}_{l,\mathrm{sp}} = \frac{1}{m}\sum_{i=1}^{m}\max\left\{d(\boldsymbol{\mu}_i, \hat{\boldsymbol{\mu}}_i^+) - d(\boldsymbol{\mu}_i, \hat{\boldsymbol{\mu}}_i^-) + mg, 0\right\} \tag{7}$$

where $\hat{\boldsymbol{\mu}}_i^+$ and $\hat{\boldsymbol{\mu}}_i^-$ represent the features of chosen positive and negative samples, respectively, which attain the minimum and maximum path distance to the anchor point $\boldsymbol{\mu}_i$ in the constructed pseudo-category-relation matrix. m denotes the number of separation pairs within one-batch samples. The function $d(\cdot)$ signifies the Euclidean distance. mg is designated as 0.5 and serves as the margin.

C. CA and OF Learning

We have three-level framework and denote the backbone network of lth level as $\mathbf{f}_\theta : \boldsymbol{\mu}_l \rightarrow y_l$ where $y_l \in \{1, \ldots, c\}$,

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where $\boldsymbol{\mu}_l$ is the learned middle feature. The gradient of Cross-Entropy loss concerning the learned feature consists of a "pull" term, which guides the feature toward its corresponding class prototype, and a "push" term, which encourages distancing from prototypes of other classes. The multilayer network structure is designed to extract more effective and diverse features from RS scene images. To this end, we propose multilayer OF constraints

$$\mathcal{L}_{l,\mathrm{og}} = \frac{1}{2}\sum_{j=1}^{l-1}\left(\hat{\boldsymbol{\mu}}_j^T \hat{\boldsymbol{\mu}}_l\right)^2 \tag{8}$$

where $\hat{\boldsymbol{\mu}}_l$ is the normalized feature by $\hat{\boldsymbol{\mu}}_l = \boldsymbol{\mu}_l / \|\boldsymbol{\mu}_l\|$. The total loss is an average over a batch of input $\boldsymbol{x}$. It can be noted that the gradient of the loss of $\mathcal{L}_{l,\mathrm{og}}$ with respect to $\hat{\boldsymbol{\mu}}_l$ is as in the following equation:

$$\frac{\partial \mathcal{L}}{\partial \hat{\boldsymbol{\mu}}_l} = \sum_{j=1}^{l-1} \cos\angle\left(\hat{\boldsymbol{\mu}}_j, \hat{\boldsymbol{\mu}}_l\right)\hat{\boldsymbol{\mu}}_j \tag{9}$$

where the gradient pulls feature $\hat{\boldsymbol{\mu}}_l$ toward the orthogonal direction of $\hat{\boldsymbol{\mu}}_j$, which is the learned prototype of the upper level branch network. The converged features will be optimized toward the orthogonal direction with $\hat{\boldsymbol{\mu}}_j$.

Given the learned last-layer feature map $\boldsymbol{F} \in \mathbb{R}^{r\times w\times h}$, where r is the number of channels and w and h are the spatial size of width and height of the feature maps. To obtain spatial information, we calculate two types of initial attention maps using average and max pooling operations across the channel dimension: $\boldsymbol{F}_{\mathrm{avg}} \in \mathbb{R}^{1\times w\times h}$ and $\boldsymbol{F}_{\mathrm{max}} \in \mathbb{R}^{1\times w\times h}$, which are subsequently concatenated and processed through a standard convolution layer to generate the 2-D spatial attention map. The final attention map $\boldsymbol{A}_l$ for the lth level branch network can be formulated as $\boldsymbol{A}_l = \sigma(\mathrm{conv2d}([\boldsymbol{F}_{l,\mathrm{avg}}, \boldsymbol{F}_{l,\mathrm{max}}]))$. We further propose a cascade spatial visual attention module, which integrates the spatial information learned by multiple branch networks and transfers the spatial information learned by upper level branch networks to the lower level branch networks. The CA can be defined as in the following equation:

$$\hat{\boldsymbol{A}}_l = \prod_{j=1}^{l} \boldsymbol{A}_j \tag{10}$$

where $\boldsymbol{A}_j \in \mathbb{R}^{w\times h}$ is the attention map calculated by the jth branch network. The learned semantic knowledge across the spatial dimension can be integrated into the lower level branch networks for enhancing the area that behaviors discriminative actor for distinguishing RS categories.

The total loss of optimizing the whole multilevel framework is composed of three parts: OF learning, hierarchical knowledge distillation, and fusion. The loss can be formulated as in the following equation:

$$\mathcal{L} = \sum_{l=1}^{L}\mathcal{L}_{l,\mathrm{cl}} + \alpha\sum_{l=1}^{L}\mathcal{L}_{l,\mathrm{sp}} + \beta\sum_{l=1}^{L}\mathcal{L}_{l,\mathrm{og}} \tag{11}$$

where L is the level number for indicating the three-level branch-networks. $\mathcal{L}_{l,\mathrm{cl}}$, $\mathcal{L}_{l,\mathrm{sp}}$, and $\mathcal{L}_{l,\mathrm{og}}$ are the calculated losses based on the hierarchical knowledge distillation, selected separation, and OF learning, respectively. α and β act as two hyper parameters for balancing the three types of losses. The whole process of training the proposed scheme is summarized in Algorithm 1.

V. Experiments

A. Experimental Settings and Datasets

In our experiments, we assess the proposed method's performance using three benchmark datasets: AID30 [38], UCMerced21 [37], and NWPU-RESISC45 [39]. Fig. 5 provides examples of images from these datasets, showing notable intra-class variations and relatively minor inter-class differences within different scenes. We begin by resizing the images and subsequently applying random cropping to achieve a resolution of 256 × 256 pixels. The standard datasets are divided into training, validation, and testing sets as established in previous studies. To improve the model's generalization abilities, we apply standard data augmentation methods, including random horizontal flips, rotations, and cropping. Two typical backbone networks of ResNet50 architecture [41] and ViT-B [42] are adopted for comprehensive comparison. We initialize both models with pretrained parameters from the ImageNet dataset [43] and fine-tune them on each RS dataset. For the classification criterion, we employ the cross-entropy loss function and utilize stochastic gradient descent (SGD) as the optimizer. We incorporate a cosine annealing schedule to facilitate convergence during the training stage.

In detail, the experimental settings are configured with a batch size of 64, an initial learning rate of 0.035, and a weight decay of 1×10^{-4}, respectively. The epoch number is 100. To ensure the reliability and reproducibility of our results, the experiment was conducted five times, and the average classification results on the test set are reported with mean and variance calculations providing valuable insights into the consistency and performance variance. Moreover, model ablation studies were conducted to comprehensively investigate the effectiveness of each proposed module, finding out their individual contributions to the overall performance improvement. These experiments are conducted on a high-performance NVIDIA RTX A6000 GPU to efficiently handle the computational demands. The network architecture is implemented using the PyTorch framework which offers a robust and flexible environment for constructing and training the models.

B. Model Ablation Studies

To evaluate the effectiveness of the proposed pseudo-category-relation learning (PCRL), hierarchical feature distillation and learning (HFDL), CA and OF learning modules, we adopt different settings for these proposed modules and analysis the experimental results. The backbone network of ResNet50 architecture [41] is adopted during the ablation studies. Experimental results on the validating set is demonstrated and analyzed to evaluate the proposed method and obtain the best settings.

1) Evaluation of Pseudo-Category-Relation and Hierarchical Learning: Experiments are conducted under different setting of the ground truth category-relation matrix D to evaluate the effectiveness of the proposed PCRL module,

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Fig. 5. Some sample images of three RS datasets. (a) Demonstrates image samples of UCMerced dataset with 21 classes. (b) Demonstrates the dataset of AID which contains 30 classes. (c) Dataset of NWPU45 which contains 45 classes.

TABLE II

PERFORMANCE OF STUDENT NETWORK TRAINED WITH DIFFERENT VALUES OF α. THE TRAINING SET IS DIVIDED INTO TWO PARTS (80%/20%) TO EVALUATE THE EFFECT OF α. GT $\boldsymbol{D}$ DENOTES THE GROUND TRUTH CATEGORY RELATIONS GENERATED FROM THE HIERARCHICAL LABELS

GT D	α	**AID30**(20%)	**UCM21**(50%)	**NWPU45**(10%)
w	0	94.41±0.12	98.63±0.11	91.28±0.15
	0.1	95.28±0.17	98.69±0.06	91.54±0.18
	0.2	95.13±0.09	98.72±0.07	91.80±0.19
	0.3	95.11±0.15	98.77±0.12	92.05±0.11
	0.4	95.26±0.11	98.81±0.08	**92.14±0.13**
	0.5	95.09±0.14	**98.82±0.10**	92.13±0.08
	0.6	**95.12±0.18**	98.79±0.13	92.08±0.17
	0.7	95.06±0.15	98.67±0.15	92.02±0.13
w/o	0	94.56±0.14	98.74±0.07	91.53±0.18
	0.1	94.78±0.19	98.96±0.13	91.91±0.07
	0.2	95.16±0.20	98.87±0.15	92.29±0.14
	0.3	95.31±0.18	**99.04±0.11**	**92.47±0.15**
	0.4	**95.33±0.14**	98.01±0.09	92.43±0.12
	0.5	95.27±0.17	98.99±0.13	92.34±0.17
	0.6	95.21±0.16	98.04±0.17	92.30±0.16
	0.7	95.13±0.18	98.97±0.15	92.16±0.11

TABLE III

PERFORMANCE OF STUDENT NETWORK TRAINED WITH DIFFERENT HIERARCHICAL TRIPLET WEIGHT β. THE TRAINING SET IS DIVIDED INTO TWO PARTS (80%/20%) TO EVALUATE THE EFFECT OF THE PARAMETERS β. GT $\boldsymbol{D}$ DENOTES THE GROUND TRUTH CATEGORY RELATIONS GENERATED FROM THE HIERARCHICAL LABELS

GT D	β	**AID30**(20%)	**UCM21**(50%)	**NWPU45**(10%)
w	0	95.09±0.06	98.82±0.09	92.13±0.10
	0.3	95.28±0.16	98.92±0.13	92.24±0.11
	0.6	95.43±0.11	99.07±0.10	92.40±0.14
	0.9	**95.51±0.12**	99.13±0.15	92.53±0.13
	1.2	95.49±0.17	99.21±0.12	92.61±0.17
	1.5	95.47±0.09	**99.27±0.11**	**92.63±0.13**
	1.8	95.39±0.13	99.14±0.15	92.55±0.16
	2.1	95.16±0.10	99.03±0.12	92.42±0.19
w/o	0	95.31±0.10	99.04±0.06	92.47±0.12
	0.3	95.38±0.15	99.11±0.09	92.54±0.17
	0.6	95.49±0.13	99.17±0.12	92.63±0.19
	0.9	95.60±0.11	**99.23±0.08**	92.75±0.16
	1.2	**95.63±0.14**	99.21±0.11	**92.81±0.14**
	2.5	95.61±0.18	99.17±0.13	92.79±0.18
	1.8	95.52±0.12	99.15±0.07	92.72±0.12
	2.1	95.49±0.14	99.09±0.14	92.63±0.17

i.e., using the binary category relation matrix D constructed according to the ground truth hierarchical labels or using the learned pseudo-category-relation matrix. As demonstrated in Table II, the proposed method runs with the ground truth hierarchical labels without relying on the proposed PCRL module by setting the parameter to w/o. In this case, we do not need to learn and construct the category relationship matrix D and use the manually annotated hierarchical label information directly to construct the soft-label $\hat{\boldsymbol{p}}$ as defined in (5). As in Table II, the separation loss does not participate in optimizing the network when α is set to 0. When α goes larger, the performance is boosted with the help of separation loss for distinguishing these hard classes.

We also showcase the results under condition of relying on the learned pseudo-category-relation matrix D. In this case, the model will perform statistical analysis on the learned features and construct a category relationship matrix. Other branch networks will further focus on those categories that are difficult to distinguish based on the constructed information. The results, as presented in Table II, reveal the impact of varying values of α on the model's performance. When α is set to 0, indicating the absence of the proposed hierarchical separation loss, we observe a baseline performance. However, as we incrementally increase α from 0 to 0.7, the proposed module consistently delivers noteworthy improvements, achieving gains of 0.85%, 0.30%, and 0.94% on the three datasets with the learned category relations, respectively. It can be noted that the learned category relations outperform the modules with the ground truth labels. This consistent trend underscores the crucial role played by the hierarchical separation loss in equipping the network with the capacity to learn more discriminative and informative features. These results not only highlight the effectiveness of the proposed HFDL module but also demonstrate its potential to enhance the model's ability to distinguish between diverse land use categories in RS scene classification.

2) Evaluation of OF Learning and CA: After obtaining the best setting for the pseudo-category-relation and hierarchical

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TABLE IV
COMPARISON OF THE MODEL'S PERFORMANCE WITH/WITHOUT USING THE PROPOSED CA MODULE. THE TRAINING SET IS DIVIDED INTO TWO PARTS (80%/20%) FOR TRAINING AND EVALUATION. GT $\boldsymbol{D}$ DENOTES THE GROUND TRUTH CATEGORY RELATIONS GENERATED FROM THE HIERARCHICAL LABELS

GT D	CA	**AID30**(20%)	**UCM21**(50%)	**NWPU45**(10%)
w	w/o	95.47±0.09	99.27±0.19	92.63±0.11
	w	**95.72±0.10**	**99.58±0.18**	**92.88±0.15**
w/o	w/o	95.63±0.09	99.21±0.16	92.81±0.17
	w	**95.96±0.15**	**99.57±0.14**	**93.17±0.12**

Learning modules, evaluation experiments are further conducted with different settings of CA and OF learning modules. In detail, we fix the value of α to 0.3 and change the β to showcase its effect to the model's performance under the scenario without ground truth $\boldsymbol{D}$. When the ground truth $\boldsymbol{D}$ is available, we fix the α to 0.5 and change the β to evaluate the model's performance. Follow the setting of the ground truth of category-relation matrix $\boldsymbol{D}$, model with/without this ground truth information are evaluated. With the increase of β, the OF learning loss $\mathcal{L}_{\text{og}}$ participates in optimizing the network by constraining the learned features to obtain more information knowledge and reduce feature redundancy. As demonstrated in Table III, the OF learning loss cooperates with the ground truth information of $\boldsymbol{D}$ and boosts the performance of the model when the β is larger than 0.3. That means the redundancy of the learned features by the three branch-networks are reduced and more effective knowledge is obtained for higher accuracy on the three datasets.

The scenario of without the ground truth category-relation matrix of $\boldsymbol{D}$ is also evaluated by learning the pseudo-category-relation matrix for the hierarchical and OF learning modules. As showcased in Table III, the increase of β boosts the performance of the classification model and achieves higher gain compared with the ground truth D. That is because the granularity of the learned category-relation information is more suitable for corresponding classification task and data. The gains achieved by the learned pseudo-category relation evaluate the effectiveness of the proposed module. Meanwhile, we also evaluate the CA module as demonstrated in Table IV. It can be found that the performance on the three RS scene datasets are improved by the CA, which focusing on informative regions of the scene images and enhancing the features within the selected spatial region for more efficient scene classification. This comprehensive evaluation allows us to quantify the impact of the OF learning and CA modules on model performance and assess its capacity to effectively leverage discriminating information, ultimately improving scene classification accuracy.

3) Evaluation of Hierarchical Labels: We further analyze the impact of constructed hierarchical information on model performance. As shown in Table V, we compared several recent hierarchical information-based methods and compared their experimental results, verifying the advantages of our method in utilizing hierarchical information and recognition accuracy. It can be noted that the hierarchical class structure is modeled by integrating this metric in the supervision of a prototypical network in Guided-Proto [45]. Similarly, a novel hierarchy preserving losses is proposed in [44] to apply a hierarchical penalty to the contrastive loss and enforce the hierarchy constraint. For TransHP [46], ancestor class is adopted as a prompt to draw focus on the subtle differences among descendant classes. Although they are not designed to recognize the RS scene images, impressive performance is achieved by addressing the hierarchical information during recognizing the images.

In addition to the overall accuracy, we also introduce the metric of average hierarchical cost (AHC) [47], [48] to better showcase the performance of the approaches regarding the hierarchy

$$\text{AHC}(y, z) = \frac{1}{|\mathcal{N}|} \sum_{n \in \mathcal{N}} D\left[y_n, z_n\right] \tag{12}$$

where $D[y_n, z_n]$ is denoted as the length of the shortest path between the nodes y_n and z_n for sample n across the tree-shaped class taxonomy. $\mathcal{N}$ is the dataset and $|\mathcal{N}|$ denotes the size of it. $D[y_n, z_n] = 0$ for all samples n satisfy $y_n = z_n$. It can be noted that our method achieves the highest accuracy and the best AHC on most experimental settings. As demonstrated in Table V, the constructed ground truth hierarchy is used to train the classification models. Compared with the baseline method that is not specifically designed to use the hierarchical labels to boost the performance of the model, these hierarchical methods improve the overall accuracy as well as lower values of AHC. That means that hierarchical information better optimizes the classification model. While improving the classification accuracy, it also reduces the hierarchical cost between the recognition result and the true category.

C. Performance Comparison

The best settings for the two parameters, α and β, have been determined in Section V-B, and we apply these settings to conduct experiments in this section. To ensure statistical significance, the experiments are repeated five times, and we report the average classification results on the respective test sets. Performance of the proposed method with and without the ground truth hierarchy are also compared and analyzed. Additionally, we provide a detailed analysis of the results, including confusion matrices, and per-category accuracy, which are presented in Figs. 6 and 7, respectively.

1) AID30 Dataset: Comparison of the experimental results are presented in Tables VI. Compared to these state-of-the-art methods, our innovative pseudo-category relation and informative feature learning framework consistently outperform them, achieving an accuracy of 95.96% and 97.58% using the ResNet50 backbone network. In addition to the ResNet50 backbone network, larger backbone network of ViT-B [42] is adopted to the teacher model. Recent advancements in the field have introduced noteworthy techniques. For instance, [52] presents a dual-branch framework designed to concurrently learn homogeneous and heterogeneous information from RS scenes, while [51] introduces a spatial-channel feature-preserving vision transform model that considers the intricate geometric details of RS imagery. Notably, both [51] and [52] achieve higher accuracy rates compared to the baseline

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TABLE V

COMPARISON OF HIERARCHICAL METHODS WHEN USING THE CONSTRUCTED HIERARCHY. † DENOTES RESNET50-BASED APPROACHES. § DENOTES VIT-B-BASED APPROACHES. ⋆ DENOTES THE USAGE OF CONSTRUCTED GROUND TRUTH HIERARCHY. OA IS THE OVERALL ACCURACY. AHC IS DEFINED AS IN (12). LOWER AHC REPRESENTS BETTER RESULTS

Method	AID30(20%)		UCM21(50%)		NWPU45(10%)	
	OA(%)	AHC	OA(%)	AHC	OA(%)	AHC
ResNet50 [41] †	92.57±0.17	0.38±0.02	98.28±0.66	0.29±0.03	88.48±0.21	0.61±0.03
HiMulConE(ResNet50) [44] †⋆	95.09±0.13	0.31±0.04	99.02±0.19	0.18±0.02	92.56±0.13	0.38±0.04
Guided-Proto(ResNet50) [45] †⋆	95.20±0.19	0.29±0.04	98.84±0.19	0.16±0.04	92.47±0.12	0.41±0.03
TransHP(ResNet50) [46] †⋆	95.31±0.17	0.27±0.05	98.96±0.21	0.17±0.02	92.51±0.18	0.39±0.05
ViT-B [42] §	95.21±0.16	0.29±0.03	98.58±0.15	0.22±0.02	92.45±0.12	0.46±0.03
HiMulConE(ViT-B) [44] §⋆	95.73±0.09	0.22±0.04	99.10±0.23	0.14±0.05	93.28±0.17	0.35±0.05
Guided-Proto(ViT-B) [45] §⋆	95.79±0.11	0.20±0.02	98.91±0.20	0.16±0.03	93.19±0.11	0.36±0.03
TransHP(ViT-B) [46] §⋆	95.93±0.14	0.21±0.03	99.07±0.13	0.12±0.02	93.32±0.19	0.31±0.04
Proposed (ResNet50) †⋆	95.72±0.14	0.21±0.02	99.56±0.16	0.12±0.01	92.87±0.14	0.33±0.04
Proposed (ViT-B) §⋆	**96.24±0.16**	**0.16±0.03**	**99.61±0.14**	**0.10±0.02**	**93.75±0.19**	**0.21±0.02**

TABLE VI

COMPARISON ON AID DATASET UNDER DIFFERENT TRAIN/TEST RATIO SETTINGS AND TWO BACKBONE NETWORKS. † DENOTES RESNET50-BASED APPROACHES. § DENOTES VIT-B-BASED APPROACHES. ⋆ DENOTES THE USAGE OF CONSTRUCTED GROUND TRUTH HIERARCHY. THE ◇ REPRESENTS THE USAGE OF LEARNED HIERARCHY

Method	AID30	
	OA(20/80%)	OA(50/50%)
ResNet50 [41] †	92.57±0.17	95.96±0.17
AGOS [49] †	94.99±0.24	97.01±0.18
EMSCNet(ResNet50) [50] †	95.13±0.10	96.96±0.10
MGSNet [18] †	95.46±0.21	97.18±0.16
ViT-B [42] §	95.21±0.16	96.96±0.22
SCViT [51] §	95.56±0.17	96.98±0.16
HHTL [52] §	95.62±0.13	96.88±0.21
EMSCNet(ViT-B) [50] §	96.02±0.18	97.35±0.17
Proposed (ResNet50) †⋆	95.72±0.14 ↑**0.26**	97.33±0.19 ↑**0.15**
Proposed (ViT-B) §⋆	96.24±0.16 ↑**0.22**	97.62±0.13 ↑**0.27**
Proposed (ResNet50) †◇	95.96±0.12 ↑**0.50**	97.58±0.15 ↑**0.40**
Proposed (ViT-B) §◇	**96.43±0.13** ↑**0.41**	**97.73±0.18** ↑**0.38**

TABLE VII

COMPARISON ON UCM21 DATASET UNDER DIFFERENT TRAIN/TEST RATIO SETTINGS AND TWO BACKBONE NETWORKS. † DENOTES RESNET50-BASED APPROACHES. § DENOTES VIT-B-BASED APPROACHES. ⋆ DENOTES THE USAGE OF CONSTRUCTED GROUND TRUTH HIERARCHY. THE ◇ REPRESENTS THE USAGE OF LEARNED HIERARCHY

Method	UCM21	
	OA(50/50%)	OA(80/20%)
ResNet50 [41] †	98.25±0.66	98.69±0.49
EMSCNet(ResNet50) [50] †	98.70±0.46	99.44±0.16
AGOS [49] †	99.24±0.22	99.71±0.25
MGSNet [18] †	-	99.76±0.14
ViT-B [42] §	98.58±0.15	99.28±0.21
SCViT [51] §	98.90±0.19	99.57±0.31
HHTL [52] §	98.87±0.28	99.48±0.28
EMSCNet(ViT-B) [50] §	99.10±0.20	99.71±0.17
Proposed (ResNet50) †⋆	99.56±0.16 ↑**0.32**	99.83±0.23 ↑**0.07**
Proposed (ViT-B) §⋆	**99.61±0.14** ↑**0.51**	**99.90±0.17** ↑**0.19**
Proposed (ResNet50) †◇	99.57±0.12 ↑**0.33**	99.86±0.11 ↑**0.10**
Proposed (ViT-B) §◇	99.59±0.16 ↑**0.49**	99.86±0.12 ↑**0.15**

model, highlighting their significant contributions to the field. Additionally, comparisons of the proposed method with and without the use of constructed hierarchical labels are provided. The results clearly demonstrate that the learned hierarchical information outperforms the self-defined hierarchy on the RS datasets. This superiority is attributed to the learned hierarchical information's adaptability and its ability to more accurately capture differences between similar categories, unlike the discrete distances defined by the relationships in the self-defined hierarchy.

2) UCM21 Dataset: As shown in Table VII, our proposed method achieves the highest accuracy of 99.57% and 99.86% on this dataset. Notably, the learned pseudo-category-relation information boosts the performance of recognizing different types of scene classes and helps the classification network achieve higher accuracy on this dataset. This allows the network to acquire more discriminative features and develop a more generalized classifier for effectively distinguishing between different scene classes. In Table VII, we expand our evaluation to encompass experiments utilizing different backbone networks, such as ViT-B [42], to comprehensively assess the effectiveness of our proposed method. Notably, when compared with the baseline model, [51] and [52] achieve higher accuracy rates. Furthermore, [50] introduces an innovative contrastive learning framework based on multiscale hard features, resulting in an exceptional accuracy of 99.59% and 99.86%. Despite these strong competitors, our method distinguishes itself by its ability to extract more informative knowledge from scene images, ultimately achieving the highest accuracy on this dataset.

3) NWPU45 Dataset: Table VIII presents the classification results of our proposed model alongside several state-of-the-art approaches. It is worth noting that the overall accuracy on this dataset is relatively lower, primarily due to the increased complexity associated with classifying 45 distinct RS categories. Recent advancements in the field have introduced innovative techniques to address this complexity. For instance, [18] introduces a multistage self-guided separation network, both achieving remarkable accuracy results. However, when compared with these state-of-the-art methods, our novel informative feature learning scheme consistently outperform them, achieving accuracy rates of 93.17% and 95.21% when using ResNet50 as the backbone architecture. This improved performance underscores the network's proficiency

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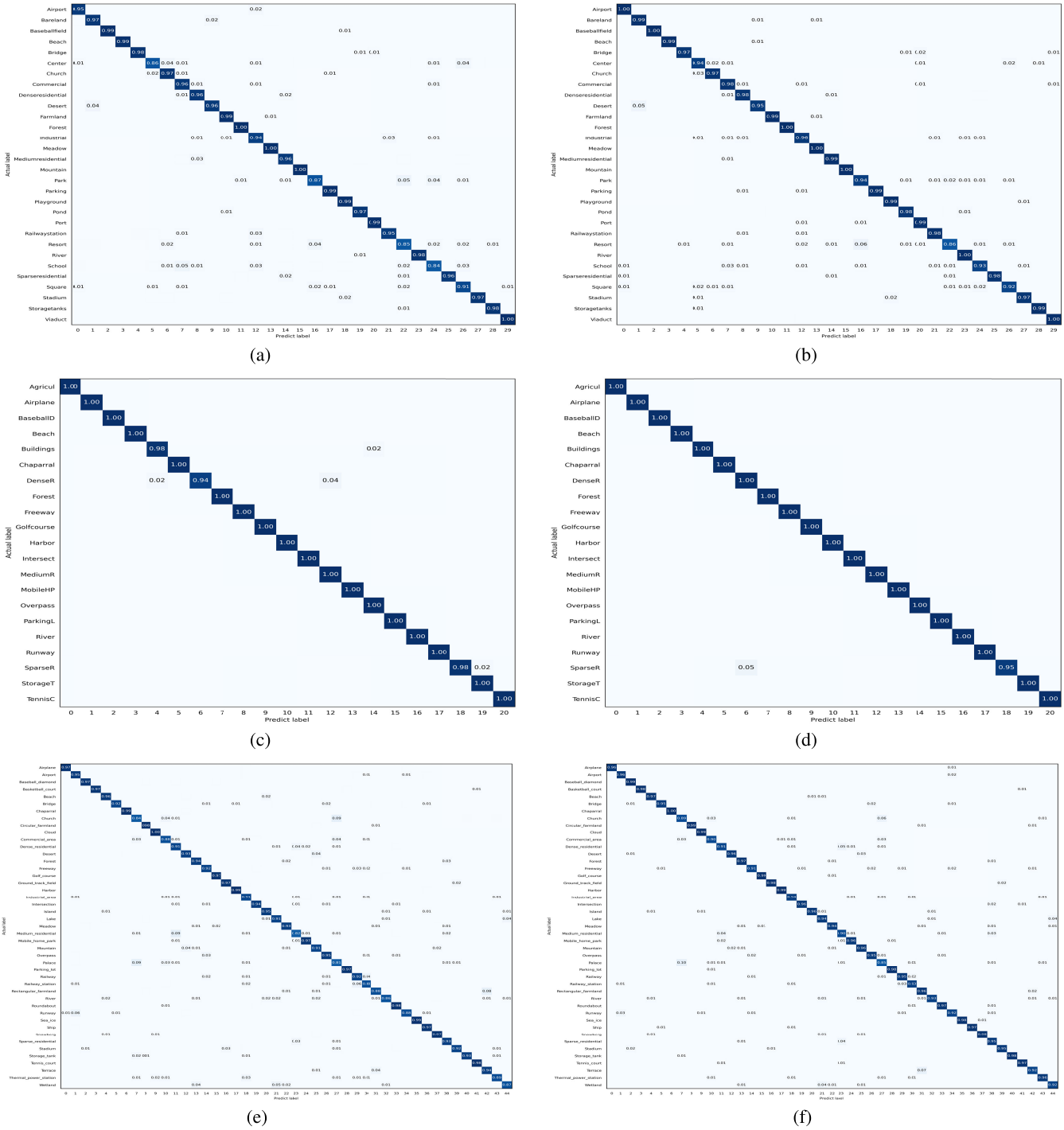


Fig. 6. Experimental results of the proposed method. (a) and (b) Are two confusion matrix obtained by the proposed model on the AID30 dataset with 20% and 50% training samples. (c) and (d) Two confusion matrix obtained by the proposed mode on the UCMerced dataset with 50% and 80% training samples. (e) and (f) Two confusion matrix obtained by the proposed model on the NWPU45 dataset with 10% and 20% training samples, respectively.

in acquiring more discriminative features, ultimately resulting in a more versatile classifier and enhancing its ability to effectively distinguish between various scene classes. We also evaluate the performance of our proposed method with a different configuration, where ViT-B [42] serves as the backbone network. In Table VIII, it is evident that [52] introduces a dual-branch framework aimed at simultaneously acquiring homogeneous and heterogeneous information from RS scenes. Similarly, [51] puts forth a spatial-channel feature-preserving vision transform model that intricately considers geometric details in RS imagery. Furthermore, [50] introduces an innovative contrastive learning framework grounded in multiscale hard features, yielding commendable accuracy results. In contrast to these approaches, our model effectively gleans valuable knowledge from the input scene images, resulting in the acquisition of more generalized and robust features for the discrimination of RS scene categories. This highlights the adaptability and proficiency of our scheme in the field of RS scene classification with diverse model configurations.

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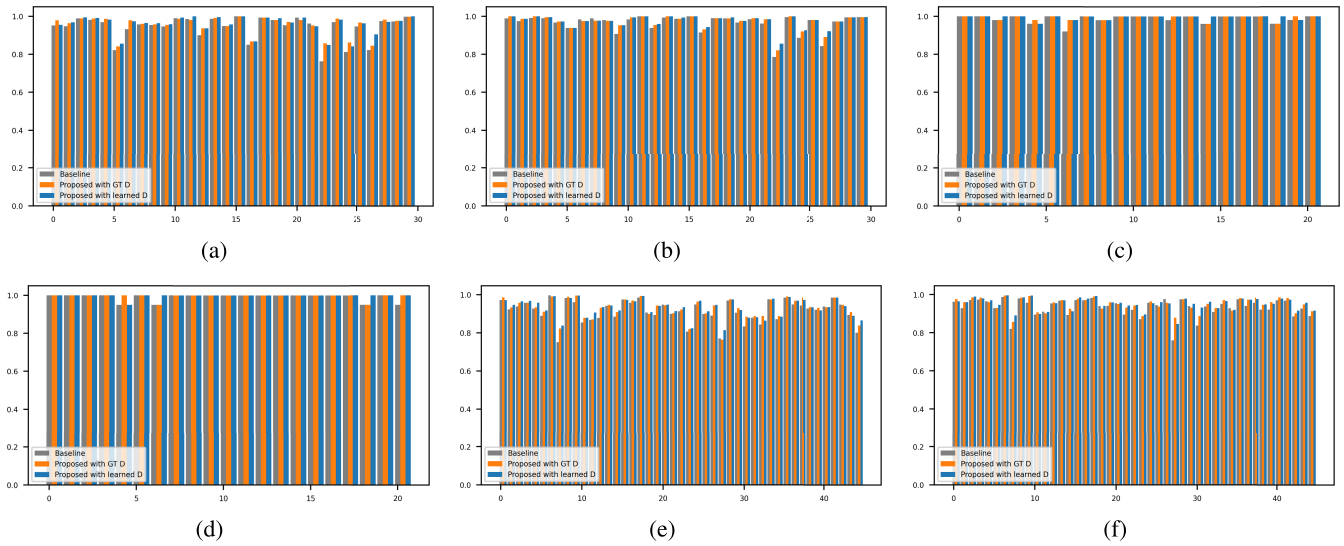


Fig. 7. Per-category accuracy obtained by the proposed model across three datasets. (a) and (b) Results on the UCMerced dataset with 50% and 80% training ratios. (c) and (d) Present the outcomes on the AID dataset with 20% and 50% training ratios. (e) and (f) Display the results on the NWPU dataset with 10% and 20% training ratios. Per-category accuracy achieved by the proposed model on the three datasets. (a) and (b) Are the results on the UCMerced dataset under 50% and 80% training ratio. (c) and (d) Are the results on the AID dataset under 20% and 50% training ratio. (e) and (f) Are the results on the NWPU dataset under 10% and 20% training ratio.

TABLE VIII

COMPARISON ON NWPU45 DATASET UNDER DIFFERENT TRAIN/TEST RATIO SETTINGS AND TWO BACKBONE NETWORKS. † DENOTES RESNET50-BASED APPROACHES. § DENOTES VIT-B-BASED APPROACHES. ⋆ DENOTES THE USAGE OF CONSTRUCTED GROUND TRUTH HIERARCHY. THE ◇ REPRESENTS THE USAGE OF LEARNED HIERARCHY

Method	NWPU45	
	OA(10/90%)	OA(20/80%)
ResNet50 [41] †	88.48±0.21	91.86±0.19
SEMSDNet [53] †	91.68±0.39	93.89±0.63
EMSCNet(ResNet50) [50] †	92.16±0.07	94.08±0.20
AGOS [49] †	92.35±0.17	94.28±0.16
LML [54] †	92.67±0.15	94.73±0.11
ViT-B [42] §	92.45±0.12	94.74±0.12
SCViT [51] §	92.72±0.04	94.66±0.10
HHTL [52] §	92.07±0.44	94.21±0.09
EMSCNet(ViT-B) [50] §	93.58±0.22	95.37±0.07
Proposed (ResNet50) †⋆	92.87±0.14 ↑0.20	94.95±0.10 ↑0.22
Proposed (ViT-B) §⋆	93.75±0.19 ↑0.17	95.61±0.13 ↑0.24
Proposed (ResNet50) †◇	93.17±0.16 ↑0.50	95.21±0.11 ↑0.48
Proposed (ViT-B) §◇	**93.92±0.14** ↑0.34	**95.83±0.08** ↑0.54

D. Report of Confusion Matrix and Per-category Accuracy

In addition to evaluating overall accuracy, we have generated confusion matrices to provide a comprehensive illustration of the model's performance, as displayed in Fig. 6. For fair comparison, we adopt ResNet50 as the backbone network. To facilitate a more intuitive assessment of our models, we also calculate per-category accuracy as depicted in Fig. 7.

For AID30 dataset, confusion matrix under two different training ratios are shown in Fig. 6(a) and (b). It can be noted that the proposed scheme achieves high classification performance even in some scene classes with high similarities, such as "Denseresidual" (96%), "Mediumresidual" (96%), and square (91%) even under the 20% training ratio. The accuracy of the "Resort" category is only 85% and 86%, which greatly affected the overall classification accuracy. Through further analysis, these scenes are composed of trees and buildings which are easily confused with each other due to the similar structures. For per-category accuracy as illustrated in Fig. 7(a) and (b), the proposed scheme with learned category relations can achieve higher accuracy than the baseline method.

As demonstrated in Fig. 6, most of the scene categories achieves over 95% accuracy to even reach 100% when the training ratio is set 50%. However, some classes of "Buildings," "Denseresidual," "Mediumresidual" are fused because they contain similar structures. Fig. 6(b) shows the performance of the proposed model when the training ratio is set to 80% and only one sample belonging to the "Denseresidual" is misclassified as "Sparseresidual." The rest testing samples are correctly classified. Fig. 7(c) and (d) also demonstrates the comparison of the per-category accuracy achieved by the proposed method.

In the NWPU-RESISC45 dataset, classes such as "church," "palace," and "commercial area" pose challenges as they contain similar ground objects like buildings, roadways, and vegetation. In these specific classes, the proposed model achieves relatively high accuracies of 89%, 85%, and 90%, respectively. As shown in Fig. 7(e) and (f), the per-category accuracy of the model with the learned category relations consistently outperforms the compared method, primarily attributed to the effectiveness of the PCRL and hierarchical feature distillation modules. This improvement is particularly noticeable in classes that are prone to misclassification, demonstrating the value of our PCRL and knowledge fusion in enhancing the model's performance.

E. Computational Complexity

To measure the computational complexity of CNN models, FLOPs, which represent the number of floating-point

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operations, are commonly employed. For training stage, the category-relation learning module brings extra computational cost with adopting the clustering module. However, the computational complexity of our proposed hierarchical model is equivalent to that of three ResNet50 models (3*4.13 GFLOPs) during the inference stage. Moreover, these three branch networks run in a parallel way, using memory in exchange for running speed. According to our actual tests, their inference speed is about 960 images with size of 224 × 224 per second.

VI. Conclusion

In this article, we present a novel hierarchical information fusion-based knowledge distillation framework, drawing inspiration from the successful knowledge transfer techniques employed in RS scene classification. To tackle this challenging task, we restructure scene classification datasets according to established land cover categorization standards, affixing three-level labels to each image. Our framework introduces a hierarchical information fusion module that explores the inherent semantic hierarchy within RS scene datasets. The acquired knowledge is then distilled to guide a student network in acquiring more informative features. Moreover, we propose a carefully designed hierarchical triplet loss that enables the student network to focus its efforts on distinguishing similar classes within the constructed hierarchy of RS scene datasets. By fusing the results of both networks, we achieve significantly improved accuracy in RS scene classification tasks. Our experimental results on three RS scene datasets reveal the efficacy of leveraging hierarchical semantic information and distilled knowledge in scene classification task.

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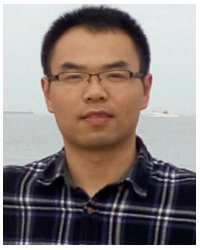
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